



UNIVERSITY OF THE PUNJAB

Fifth Semester – 2019

Examination: B.S. 4 Years Program

Roll No. in Fig.

Roll No. in Words.

PAPER: Design and Analysis of Experiments (Theory)
Course Code: STAT-303 Part-I (Compulsory)

MAX. TIME: 15 Min.
MAX. MARKS: 10

Signature of Supdt.:



Attempt this Paper on this Question Sheet only.

Please encircle the correct option. Division of marks is given in front of each question.

This Paper will be collected back after expiry of time limit mentioned above.

Q.1. Encircle the right answer, cutting and overwriting is not allowed. (1x10=10)

- i) The basic principles of experimental designs consist of:
a) Randomization b) Replication c) Local Control d) All of these
- ii) The smallest subdivision of the experimental material is called:
a) Treatments b) Experimental Unit c) Experimental Error d) None of these
- iii) In a completely randomized design, treatments are assigned to experimental units at random.
a) Completely b) Partially c) Systematically d) None of these
- iv) The assumptions under analysis of variance consist of:
a) Normality and Independence c) Linearity and Additivity
b) Both (a) and (b) d) None of these
- v) The following design provides the maximum number of degrees of freedom for error sum of squares:
a) Completely Randomized Design c) Completely Randomized Block Design
b) Latin Square Design d) None of these
- vi) Multiple comparisons tests are applicable when:
a) Null Hypothesis about equality of means is rejected
b) Null Hypothesis about equality of means is accepted
c) Does not depend upon the rejection or acceptance of Null Hypothesis
d) None of these
- vii) One can estimate the missing observation through covariance technique by simply changing the sign of....
a) b b) r c) Correction Factor d) None of these
- viii) The efficiency of two experimental designs can simply be measured through of error variances.
a) Addition b) Subtraction c) Multiplication d) Ratio
- ix) Two Latin squares are if each letter of one square design occurs exactly once with every letter of the other square when they are superimposed.
a) Orthogonal b) Factorial Designs c) Efficient d) None of these
- x) A contrast iscombination of treatments.
a) Linear b) Exponential c) Quadratic d) None of these



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PAPER: Design and Analysis of Experiments (Theory)

Course Code: STAT-303 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q2. Differentiate between the following terms:

(5x4=20)

- Fixed Effects and Random Effects
- Analysis of Variance and Covariance
- Completely Randomized and Completely Randomized Block Designs
- Replication and Randomization
- Latin Square and Graeco Latin Square Designs.

- Q3. a) In an experiment 'k' treatments and 'r' blocks are selected at random from a large number of treatments and blocks. Develop expected mean squares by clearly indicating the assumptions used.
- b) Given the following ANOVA for a CR design for four treatments: (6+4)

S.O.V	d.f.	SS
Treatments	3	1.1986
Error	36	1.0323

Test the significance of difference between treatment means by using Duncan's Multiple Range Test when treatment means for four treatments were 1.464, 1.195, 1.325, and 1.66.

- Q4. a) Seven treatments arranged in six randomized complete blocks gave the following sum of squares and products:

S.O.V	XY	YY	XX
Blocks	-111.65	6.31	7472.6
Treatments	3598.05	112.86	116020.3
Error	682.20	23.23	28665.1

- Is the regression of Y on X significant at 0.05 level of significance.
 - Construct ANOVA and write the inference.
- b) The analysis of Variance for a RCB design produced the table shown below: (6+4)

S.O.V	d.f.	SS	MS	F-Ratio
Treatments	3	28.2	-	-
Blocks	5	-	13.80	-
Error	-	34.1	-	-

Complete the ANOVA table and test the significance of difference among the treatment means.

- Q5. a) Derive formula for estimating N missing observations in a Latin Square Design when values are missing in different columns, different rows and different treatments.
- b) In an experiment to examine the effects of row spacing on the yield of wheat, 8 row spacing were used and 6 blocks of an experiment were used. The sum of squares for Total, Blocks and Treatments were 2195.48, 617.86 and 1283.65 respectively. Find the relative efficiency of this design with the design in which blocks are ignored. (7+3)